

A Beginner's Guide To How Docker Works

Understanding Docker is essential because it is the foundation of modern cloud deployment and containerization. Mastering these basics will save you hours of configuration headaches on your development journey.

What you'll learn:

- 1 Pulling Your First Image
- 2 Running Your First Container
- 3 Building Images with a Dockerfile
- 4 Persisting Data with Volumes

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Pulling Your First Image

Before you can run anything in Docker, you need to download an image from Docker Hub. Think of an image as a frozen snapshot of an operating system and your application code. When you run a pull command, Docker fetches this blueprint layer by layer to your local machine so it is ready to be turned into a running container.

```
$ docker pull ubuntu:latest
```

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Running Your First Container

Once you have an image, you can bring it to life by running it as a container. Containers are lightweight and run in their own isolated environment, meaning they won't mess with your laptop's files. The run command creates this isolated space, starts the application, and lets you interact with it immediately.

```
$ docker run -it ubuntu:latest bash
```

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Building Images with a Dockerfile

Instead of using pre-made images, you often want to package your own code. You do this by writing a simple text file called a Dockerfile. Each line in this file adds a new layer to your image, such as copying your code or installing python, creating a repeatable setup process.

```
FROM python:3.9
WORKDIR /app
COPY . .
CMD ["python", "app.py"]
```

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Persisting Data with Volumes

Containers are designed to be temporary, meaning anything created inside them is deleted when they stop. To keep your database or log files safe, you use volumes. Volumes mount a folder from your host computer directly into the container, keeping your precious data secure outside the container's lifecycle.

```
$ docker run -v /my/host/path:/app/data -d my-database-image
```


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Key Takeaways

- + Always clean up unused containers using `system prune` to save disk space.
- + Keep your Dockerfiles simple by combining `RUN` commands where possible.
- + Never store passwords or API keys directly inside your Docker images.
- + Use specific version tags like `python:3.9` instead of just `latest`.
- + Test your containers locally before pushing them to any remote registry.

Watch Out For

- ! Forgetting to map ports and wondering why the app isn't reachable in the browser.
- ! Storing critical data inside the container without attaching a volume.
- ! Making images bloated by installing unnecessary tools and packages.

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